

eSTUDENT SERVICES

A PORTAL SOLUTION

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Introduction

This document describes student services online and the use of related technology in higher education. Integrating technology into the delivery of student services is not a simple process for such a complex organization as a college. There are many things to be considered, such as: information technology (IT), network infrastructure, software, integration, and training. The key question, nonetheless, centers around the value of existing student services and how web-based initiatives could add value and help the college gain a competitive advantage. This question has its roots in business models, but is an increasingly useful framework for educational leaders making these decisions for their institution. The aim of any new design is to more effectively and efficiently help students in their quest to be successful learners, to complement traditional student services offerings and, ultimately, to adapt to change.

This document also provides a specification of the eStudent Services functionality as well as an overview of the necessary architecture. It is targeted to an audience of higher education administrators, student services administrators, academic affairs administrators, faculty, students, and technical personnel who are familiar with Internet and Enterprise Management Technologies. The reader should come away with an understanding of the functions of eStudent Services and Portal Technology, how they work, and how they can be used at institutions of higher education.

It is the opinion of this group that it is essential to be familiar with the implications of Internet and technology trends on student services.

Needs Analysis

There are several institutional and student drivers that are transforming institutions of higher education to become learner –centered and to develop and implement new models for integrated, personalized, online student services:

Institutional drivers for change

Due to declining budgets, institutions are exploring ways to automate routine services to make them more accessible to all students. With the increase in online student enrollment, institutions will be more accountable in providing student services for both online learners and on campus students. Those institutions that will use new technologies and the web to automate and integrate student services to provide learner centered services will more likely to have a competitive advantage in the higher education arena.

Students needs

The student demographics on today's community college campuses have shifted. The typical student today is not an 18 years white, middle class, male, but rather students in community colleges are more than half over 25 years of age, working adults who attend part-time and ¼ have a different ethnicity. These technologically savvy students, the "Generation Y Millennial" aptly described in the article, "Exploring the Future of Community College and Adult Learner Student Needs and Issues by Barry Hubbard, have grown up with computers and the internet and are accustomed to the self-service mode that is available to many online commercial services. The web has also accustomed them to get "just in time" information and service "by googling" when they need it. They also expect personalized service and being very diverse (first generation, ESL, international, disabled, working adult distance learner), they expect institutions to deliver services that are customized to their specific needs. Students will also choose convenient services that are easy to access 24/7.

Design Analysis

Traditional on campus Student Services

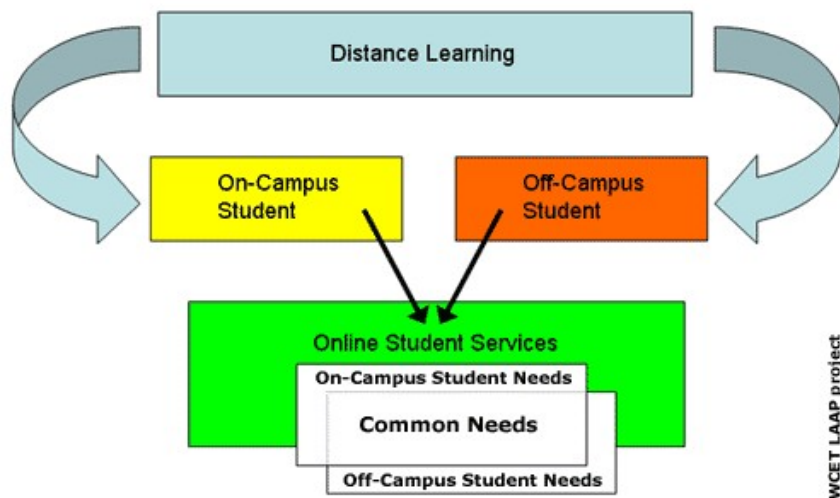
The traditional model for student services in a physical location is organized by function, with each department focusing on a specific student service area. In this traditional model students often go from one office to another, frequently waiting in line to get answers to questions or complete a form.

"Traditional" functional areas for student services that are relatively uniform across institutions are:

- Academic advising
- Admissions
- Assessment and testing
- Career planning
- Counseling
- Student Orientation
- Student information systems
- Tutoring

Over the past several years there has been a growing use of Internet technologies to serve students both off and on campus and many Colleges have recognized the need to develop Web-based, anywhere, anytime, access to traditional student services.

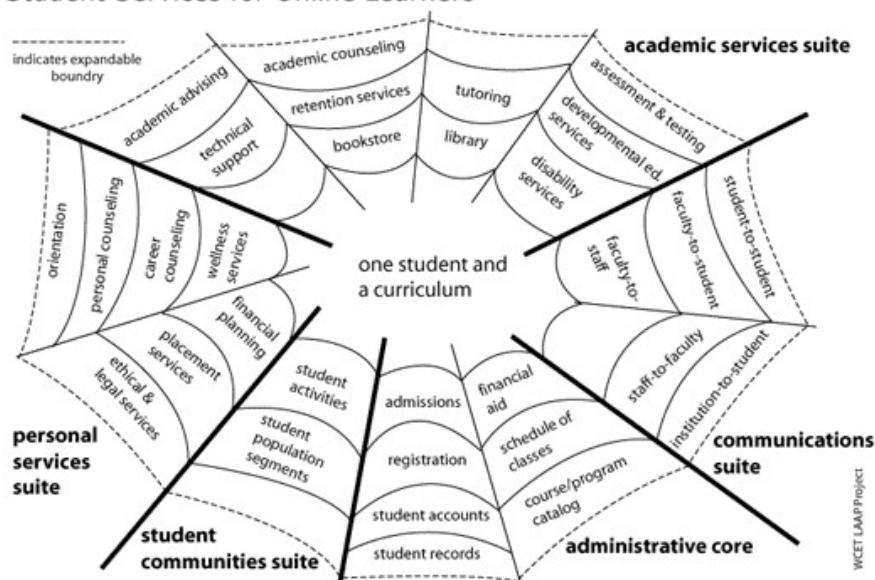
Recently, colleges have realized that online services that were developed specifically to serve distance learners have been requested and widely used by on-campus learners. So there seems to be a consensus that both off campus and on campus students have some common needs and that developing online student services will benefit both (WCET LAPP Project on Student Services 2000 – 2002)



What services should be offered online?

The graphic shows an array of student services that should be delivered online to online learners (Shea and Blakely, 2002).

Student Services for Online Learners



According to Darlene Burnett, traditional Student services are undergoing a transformation due to external trends, limitations of the traditional student services model, and increasing demands for distance learner support. She explains how the new models are student-centered and focus on added value and relationship building with every service transaction.

These new models are moving from transaction based, to experienced based, and provide for self-service and Web-based student service portals that provide decision support, personalized communications, enhanced community and process orientation.

She describes how institutions that put student services on the web move through stages from just putting static information about the services on the web to providing interactive personalized and customized services via the web (the Portal solution). She has identified 4 generations of online student services:

Generation 1 Content

The information is presented from the institution's point of view. It is organized to mirror the physical organization and processes of the institution.

Generation 2 Content in Context

The information is channeled to certain population segments. For example there are several paths for prospective and matriculating students to various student services. The services are not integrated; they still reflect the separateness "silos" of the different physical campus offices.

Generation 3 Customization, Personalization and Community

These are what Burnett refers to "one stop" student services. What distinguishes these services is that they are now aggregated and integrated with an array of related services to provide a personalized and customized service for the individual student. These services are designed from the student's point of view.

Portal solutions, such as the Black Board Community Systems, part of the Black Board Academic Suite, are providing functionalities such as modules/channels, role based delivery of information and communication tools that will enhance the student's experience and provide a truly networked campus-wide learning community environment in which personalized and customized content modules and services are delivered to different campus constituencies (role-based) such as students, faculty, staff, and administrators.

EXAMPLE 1 Blackboard Community System Demo

<http://www.blackboard.com/products/as/communitysys/>

Click on button Community System Product Demo

EXAMPLE 2: eLion

An online system to assist student to make a better decision

http://www.our.psu.edu/eliondemo/student/demo_student_page.html

At Penn State they have developed an online service to late drop a course. The student goes through several interactive steps to assess the impact of late dropping a course on his GPA, receives information on tutoring, learns about the impact of dropping a course on his financial aid and finds out how this will affect his program completion.

This system enables the blending of different services: academic advising, registration, and tutoring and financial aid information.

Generation 4 High Tech /High Touch

Services are designed to nurture a relationship between the student and the institution. Some of the new features will be using systems that have functionalities to assist students in making decisions, and that feature proactive communication tools with an array of VOIP web conferencing technologies that will allow for real time, human (albeit not f-t-f) interaction with the institution.

The technology will certainly play a key role in this evolution of online services, but the focus is still on best practices to provide student, staff and faculty-centered services with an emphasis on productivity, effectiveness, accountability, satisfaction and improved access and quality.

New Vision of Student Services

All institutions of higher education have student services, but not every student recognizes the importance of these services and they don't take advantage of them which is why they seem to either: not know of their existence, find it hard to ask for help or know where to start asking for help. For whatever reason, students are finding it difficult to access these services and it may be due to their work, schedule, family, finances, or other responsibilities. Student services are very important to student success and are about providing help. The new vision is one that brings together 24/7 services into a "one-stop helpdesk."

The challenge for institutions of higher education is bringing student services to students by making them accessible and available to students wherever they are; a blend of person-to-person interaction that's accessible from their own home, or from any remote location. One way of making these services available is by providing convenient access via the Internet, which is becoming a mean of communication in the society of today. eStudent Services therefore focuses on the use of information technology in delivering online student services.

The Internet enables the following types of communication: student-to-student, student-to-faculty, faculty-to-student, and student-to-student services. The same Internet backbone is being used by businesses to interconnect with their employees, partners, and customers. Several examples of this communication type include: Internet email access, web surfing, electronic portfolios, and blogs-- basically any type of communication device such as computer, PDA, cell phone, television that is widely accepted as a means of communication. These tools are not any longer limited to businesses, but instead, anyone with the Internet access can obtain a free email address from many providers like yahoo, msn, and google, for example. With the advances of the Internet, institutions of higher education can offer online student services to their students, faculty and employees. A new vision is to offer comprehensive online student services through a web portal.

A portal is a single Web entry point for the University community, serving as an interface to applications that are customized and authenticated to an individual user. This interface will provide access to online resources used in day-to-day transactions.

Portal Technology

The portal technology is a newly emerged web-based architecture, and it is a key component of organizing and delivering online student Services. A portal is a single Web entry point for the institution, serving as an interface to student services that are customized and authenticated to an individual user and includes a collection of resources and applications available to students, faculties, and employees. Such resources are a collection of web pages and applications that are referenced as URLs (Universal Resource Locators) links. In addition, a web portal is a WYSIWYG (what you see is what you get) graphical user interface (GUI) and is a term describing the user interaction. The online application of student services is a web portal that is accessible via a Internet browser (Internet Explorer or Netscape).

A URL is a starting point of a web portal, web page, or website and it may contain other embedded URLs as link to other resources on the web portal. The web portal is also available on the wide area network (WAN) and the local area network (LAN) as an intranet that include URLs to web pages or online applications.

A typical web portal configuration of online student services is a URL connecting to institutional resources, external resources, and personalized services. Personalized services are online resources that use role-based configurations specific to a users that include student, faculty, administrator, counselor, advisor, employee, etc. The web portal is the Internet-based technology that is composed of both the Internet technology and online applications. A more user-friendly term for a web portal is a “collection of personalized online resources tailored to individual user needs and their role at the institution.”

Traditionally, the term “student services” has described a type of services available to students at institutions of higher education that include, but are not limited to: academic advising, admissions, assessment and testing, career planning, counseling, orientation, student information systems, and tutoring. The purpose of these services has focused on assisting student with their academic study.

The ideal web portal of student services is a generation 4 (called *MyOhlone*), which features customization, personalization, and community but also allows for nurturing a relationship between the student and the institution [1]. Located at <http://myohlone.ohlone.edu>, it will be the virtual gateway for student services functions. It will be the access point for the College’s administrative information systems. This document is a paper design of eStudent Services, and as such is limited to student services and its roles and services. A vision of a completed system includes a tightly integrated system that combines institutional functions and processes with student services. There is also an information technology part that is required but it’s not addressed in this design. This specification, instead, describes the basic functionality of online student services and a capability that is scalable to future ERP solutions for higher education.

Portal Design Goal:

A key assumption in the web portal technology design is that the content will come from both internal and external sources.

The integration of the enterprise portal technology provides students, faculty, administrators, and employees with both the tools needed to build and manage a portal with multiple content sources, as well as all of the facilities needed to build and manage institutional resources and cross-enterprise system integration between content sources.

Enterprise Resource Planning

In most organizations today, technology infrastructures have become highly complex and difficult to manage, with significant overlap of systems and applications, further complicated by the fact that these systems are not well integrated. There is a clear need for a standardized enterprise architecture as a framework to help organizations make better decisions and enable them to leverage existing technology investments. Enterprise Resource Planning (ERP) is a software architecture that can quickly and effectively incorporate institutional operations into real-time everyday functions.

The information technology, Internet technology, web-based technology, and relational database management system are key technologies for ERP software solution and online student service. The information technology is a network system. The Internet technology is a public network system and protocols that allows information sharing. The web-based technology is the Internet browser technology that interprets and renders HTML (Hypertext Markup Language), which is an authoring language, used to create documents (or web pages). The

HTTP (Hypertext Markup Language) is an application protocol for distributed, collaborative, hypermedia information systems. An example of a typical URL is “<http://www.ohlone.edu/MyOhlone/StudentServices>” and there are three parts of this URL. The first part is a protocol (*http*). The second part is a domain name—server name (*ohlone.edu*). The third part is a starting point of the student services application service (*StudentServices*).

ERP is a software solution integrating information and business processes to enable sharing throughout an organization of information. It's a cross-functional and enterprise wide. The term enterprise resource planning is also management information systems that integrate and automate many of the business practices associated with the operations. Over the years with the advances of technology, Internet, and especially with the ERP software solutions, it has become widely accepted by the industry as a software solution for connecting businesses and people together through the information sharing network, the primary purpose of which is to gain a competitive edge. Such applications, in this view, are business processes and functions that are supporting and automating the business processes that include manufacturing, distribution, personnel, project management, payroll, and financials. Moreover, the business applications delivered via the Internet have also grown very rapidly, due in part to consolidation of industry by such actions as: maximizing efficiency, collaborating, improving accountability, and reducing overall operating costs.

The ERP software solution can also be implemented to help institutions of higher education to improve communication and collaboration between departments, administration, employees, faculty, and students; All functions that are involved in operations or production are integrated into one system. At the administration level, it helps collaboration between: the president and his/her staff, admissions to budgeting, budgeting to accounting, accounting to staffs/students, and campus-wide, department to department. ERP software solution will allow institutions to increase efficiency, reduce operating costs, while improving collaboration, and accountability. In essence, it makes possible the communication between all levels of the institution. It is a completed project management system tailored to individual educational needs.

Although, this design covers only student services, there are many potential ERP software solutions for institutions of higher education that could be a module of the higher education ERP software solution. The complete solution of higher education is a more tightly integrated with institutional functions and processes. For example, one such solution is a college president's web portal which oversees the whole institution from managing employees, admissions, finance, budget, etc. A faculty's web portal might be tailored to viewing his/her class activities, such as lesson plans, assignments, student communications, and other administrative tasks. A student's web portal might be for viewing his/her class schedule, assignments, advising appointments, admissions needs, assessments and testing schedules, career advice, counseling, orientations, tutoring and other activities. This design proposes a student's web portal of student services. The web portal is

a software solution delivered over the Internet and is a module of the complete solution of higher education ERP.

Higher Education ERP

ERP solution for higher education is not new. In fact, there are many solution providers that have offered comprehensive software solutions to higher education, including student services modules. Its management information systems are aimed at automating and improving processes and are tightly integrated with higher education functions and processes. The three major solution providers include Oracle, PeopleSoft, and SAP.

- Oracle Corporation, e-Business Suite for higher education

Oracle solution is comprehensive, integrated set of applications for the higher education market that automates and streamlines, every facet of campus operations and administrations, financial aid, grant management, human resources, procurement, recruiting, admissions, financial aid, grants management, project management, regulatory compliance, portal development, and communication (integrated e-mail, voice-mail, fax, scheduling, and calendar).

- SAP, Solution for Higher Education

SAP solution is a campus management that is a student information system for managing the student's "lifecycle" from admissions to graduation that include management, budgeting, resource control, learning solution, lifelong learning, distance learning, curriculum management system, business intelligence, reporting, data-mining, strategic enterprise management, enterprise portal (student portal, university, employee portal). SAP Customer Relationship Management (CRM) solution include alumni relationship management, call center, and web access.

- PeopleSoft, Higher Education Solution

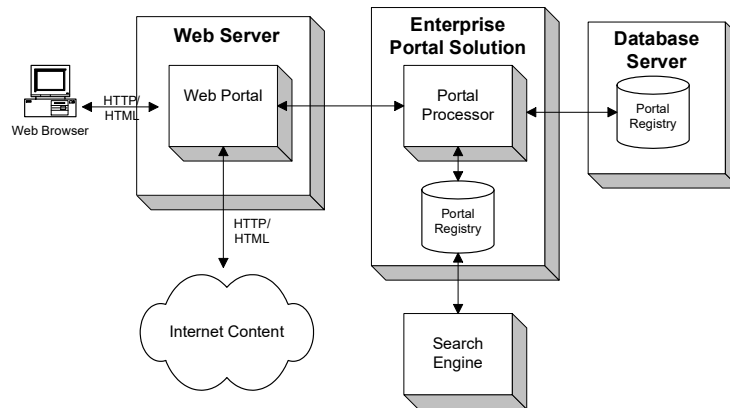
PeopleSoft solution is a roles-based component that includes Student Administration, Human Capital Management, Contributor Relations, Customer Relationship Management, Financial Management, Supplier Relationship Management, Grants Management, Asses Management, Real Estate, and Analytics. Within the student administration suite are solutions for creating a campus community, recruiting, admissions, academic advising, student records, and financial aid.

Technical Requirements

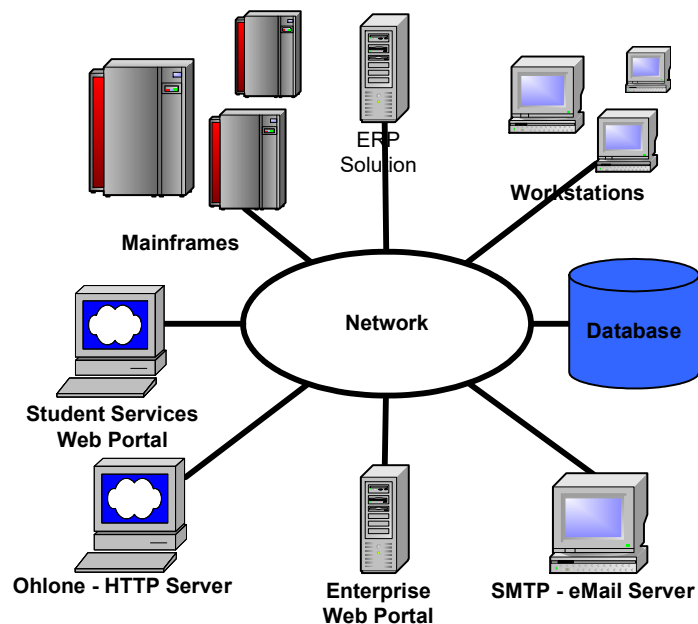
The eStudent Services and Web portal are a tightly integrated system that include the following:

- Relational database (Microsoft, DB2, Mainframe, Oracle)
- Oriented programming language
- Computer aided software engineering tools in development
- Client/server architecture,
- Open system portability
- Enterprise Portal Solution
- Enterprise Resource Planning

A typical Portal architecture



A typical network configuration



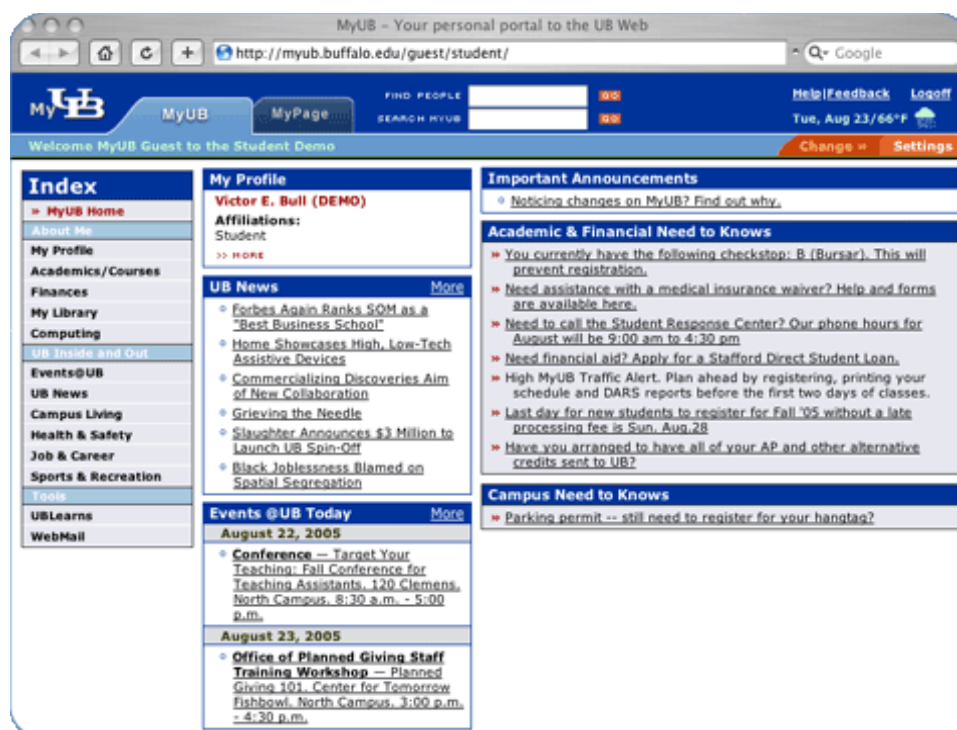
Testimonials

The market for ERP in higher education is growing. There are numerous institutions of higher education have already adopted a web-based solution for their institution. Here is a list of institutions of higher education using web portal technology.

1. San Jose State University,
https://my.sjsu.edu/psp/cmsportal/EMPLOYEE/EMPL/h/?tab=PAPP_GUEST
2. University of Washington,
<http://www.washington.edu/protos/myuw/demo/student.html>
3. University of California, Los Angeles, <http://my.ucla.edu/welcome.asp>

4. University of Buffalo, <http://www.buffalo.edu/aboutmyub/>
5. Cal State East Bay, http://www.csuhayward.edu/about_csuh/university_life/student_services.html
6. California State University, Fresno, <http://peoplesoft.csufresno.edu/portal/default.asp>
7. Syracuse University, <http://myslice.syr.edu>
8. Princeton University, <http://web.princeton.edu/sites/oit/studentservices.htm>
9. The University of Florida, <http://www.registrar.ufl.edu/admin/itac/peoplesoft.doc>
10. University of massachusetts Lowell, <http://intranet.uml.edu/peoplesoft/portal/>
11. The University of Vermont, <http://www.uvm.edu/cit/IT-news/200409/?Page=planning/Portal.html&SM=planning/planmenu.html>

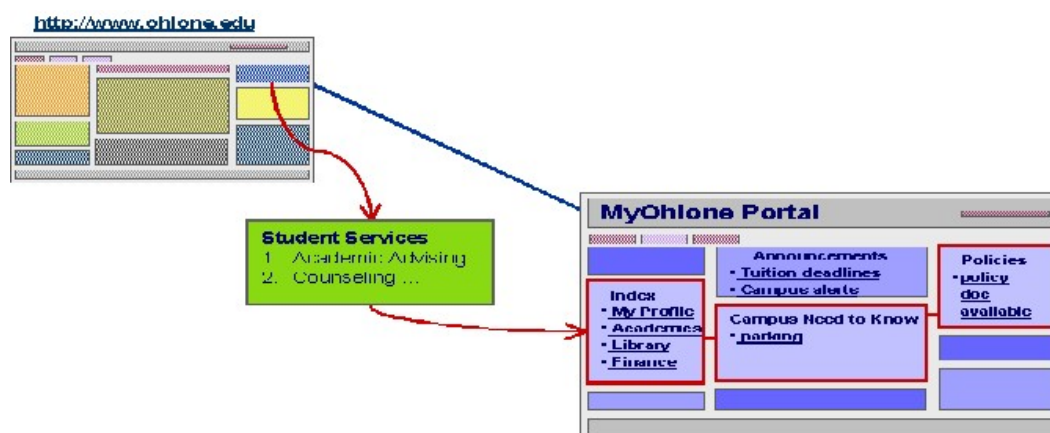
Example: University of Buffalo - Student Portal



User Portal—Personalized Home Page

The term “Portal” is a technology term for describing application that runs via the Internet browser. The portal application is a collection of applications organizing

for various roles of users. For eStudent Services portal is an organizing Internet application for students, faculty, and administrators of higher education. The following portal anatomy chart explains several of the possible references to the "portal" term. Here is an example of a portal anatomy:



Terminology

Portal Student Services – an application that provides collaborative and interactive the student services via the Internet browser.

Portal users – a web-based (online) content that is personalized to the specific user's educational needs. The portal solution is usually developed by role and targeted to a specific group of end users such as *student*, *faculty*, and *administrator*.

Portal application – an application layer retrieves and integrates the numerous student services of content (application) for the various roles accessing the portal that applications include academic advising, admissions, assessment and testing, career planning, counseling, orientation, student information systems, and tutoring.

Portal Content – application content appears within the portal. This content is independent of the portal technology that displays and organizes the content and it's designed differently for student, faculty, and administrator accessing the portal. The content of eStudent Services include *academic advising*, *admissions*, *assessment and testing*, *career planning*, *counseling*, *orientation*, *student information systems*, and *tutoring*.

ERP – Enterprise Resource Planning, it's a business management system. The ERP methodology is widely accepted by the software industry as a standardized methodology of software application for businesses. The ERP for eStudent Services application can be defined as an application suite of helping student, faculty, and administration collaboratively interconnected with each other by their roles and functions.

The following gives a brief overview of eStudent Services solutions so that the reader may gain a better understanding of how this new technology is being used in higher education.

eStudent Services Components

Components of eStudent Services include Advising, Admissions, Assessment & Testing, Career, Counseling, Orientation, Tutoring, and Student Information Systems. These components are plug-in modules of eStudent Services, an online application and are an independent application component that is personalized according to their roles accessing the portal.

Student Portal

A student portal is an online web-based resource that includes student class schedules, personal library, student business accounts, faculty contact information, email, and institutional news. When a student signs in—using the same login assigned them upon registration—the system will recognize him or her, and give information that is relevant to individualized program, campus or concerns.

Faculty Portal

When a faculty signs into the portal, the system will also recognize him or her and present news and online information relevant to the campus, program and work needs. A faculty portal will include individual class schedule and student rosters, library and teaching resources—all available with a single sign-on. One sign-on will similarly take staff members into their email. (Example: Blackboard, <http://online.alliant.edu/>).

Administrator Portal

When an administrator signs into the portal, the system will likewise recognized him or her, and present relevant information. It is very similar to the Faculty and Student portal.

Design Considerations

Some questions to consider in the process of design are as follows:

- What is eStudent Services?
- What is the value proposition of eStudent Services solution?

- What is eStudent Services Portal?
- Why is Student Services technology ‘next generation’?
- What are the key benefits of eStudent Services?
- How does eStudent Services technology improve institutions collaboration?
- What eStudent Services applications do we have?
- What are the early limitations of eStudent Services technology?

A use case is found in the following article, summarized below:

[“Group Plans New Services for Distance Education”](#)

The Southern Regional Education Board, in 2003, launched a project to “help distance education student enroll in Web-based courses and receive academic support” (Carnevale, 2003). The idea was to offer “centralized online assistance” similar to those already being provided.” Originally based on a small-scale grant, the “Ways in Mentor” is funded by no state money, but instead “companies and agencies in the financial-aid industry that hope to offer loans to students using the program.” It was also developed as a response to the need of adult learners, many of who populate online courses from multiple campuses. For example, the program will help students keep track of their course credits by offering electronic “passports” to move between colleges without having to go through long application processes, and will “keep centralized records of which courses students have completed.”

The premise of this program is that most online students are returning students, and that they “often need more help from student services than their younger classmates” as they’ve been away from it “for a period of time” (Carnevale, 2003). Indeed, colleges need to creatively “engage an increasingly digitally enabled student body” (Sifferlen, 2003). The tricky part for the SREB has been “making it work with 8,000 courses, 350 institutions, and 300 degrees across 16 states (Carnevale, 2003). The goal, nonetheless, is similar to one that this group has put forward in the portal model: a “one-stop shop” for the student from planning to graduation.

eStudent Services Functionalities

The previous section gave you an overview of eStudent Services applications that will leverage portal technology. This section describes specific functions of eStudent Service, an online student services application. Components of eStudent Services include, but are not limited to: academic advising, admissions, assessment and testing, career planning, counseling, orientation, student information systems, and tutoring. These are few interactive solution of the

unlimited possibility for comprehensive software solution with enterprise web portal.

Portal Common Components

Personalization

Personalization is much more than just allowing the end user to modify and organize their personal home page (like Ask Alliant! or Yahoo! consumer portals). With an eStudent Services portal, it is critical that the user (student, faculty, and administrator) personalizes the tool specifically for their roles. In addition, users should be able to customize their personal home page through the portal. This is one of many personalization features of the portal.

Content Management

Another key function provided by a portal is content management. Content management addresses the creation, collaboration, versioning, and publishing of content. Examples of content management features in the eStudent Services Portal technology include the facilities for managing the student advising, counseling, tutoring content categories, for example.

Through the portal content management, users can define, describe, and categorize content that is displayed through the portal. Hierarchies of content are created through these portal utility tools that are used to drive user navigation.

Search

The ideal portal uses a search engine to quickly search for registered content in the portal registry. This is a popular means of portal navigation. Google (Verity, Excite), the industry leading search engine, is packaged with portal technology.

Academic Advising

The academic advisor helps students create a plan of study to reach his academic goals. The plan must be academically customized for the specific student. Traditionally, academic advising has been viewed as a person-to-person activity. Advisors also help students feel part of the campus community, develop academic and career goals, and must understand the type of student they are serving and guide them to become successful learners.

Recently, a number of institutions are using web based technologies and communications tools to deliver online academic advising that is personalized and customized to the needs of the individual student.

According to the WICHE and NACADA guidelines for developing an advisor's web site (Wellborn, 1998 and Carnevale, 2000), the basis for a comprehensive advising site should include the following elements:

1. A clear and concise explanation of core curriculum (or general education) requirements. Making this information available online will free up some time, will give students greater control and responsibility for the advisement process.
2. A Frequently Asked Questions (FAQ) section. Every advisor spends a portion of the day repeating answers to the same questions. Putting answers to FAQs online saves time for staff and gives students access to answers as needed.
3. Informational pages for special populations/self help assistance. While there are certain common needs among students, there are segments of the population with additional unique concerns. Freshmen, students without declared majors, students on academic probation, and commuters are examples of groups with additional needs for support. Examples of information to include for advisement: career/major information, study skills building worksheets, an explanation of the academic standing policy, information on how to get computer access and technical support.
4. Links to related university sites. Holistic advisement involves supporting a student both academically and personally. Links to campus services such as student activities calendars, campus organization pages, career services are needed to make sure all needs are being met.
5. One-on-one access to advisors. To generate a more personal environment and provide opportunities for interaction, advisors are experimenting with various forms of electronic communications.. Access to a qualified advisor can be achieved through the use of chat rooms, list serves, and email.

(Excerpt from the Article "Virtual Advising Delivering Student Services" by Linda Wagner)

<http://www.westga.edu/~distance/ojdla/fall43/wagner43.html>

Effective advising online should be convenient and accessible any time, any place.

Examples of Online Academic Advising

1) e-Lion

A portal environment for Penn State students to manage the course load online

(Click on Drop link in the left column and follow pages to see a session of online academic advising)

2) On Course Home Page, University of Arizona

<http://wacky.ccit.arizona.edu/~oncourse/home3.html>

On Course is a computerized degree audit/advising support system designed to help students achieve their academic goals efficiently.

3) Ohlone College Online Counselor/Advisor

<http://www.ohlone.edu/org/counseling/electcounsel.html>

Students can fill out a form and ask questions to an electronic advisor/counselor .response time is 24/ 48 hours

[Ohlone College Online Counselor](#)

NACADA National Academic Advising Association

[Distance Learning, by George Steele](#)

Admissions

[The Personal Touch Is Gone From College Admissions](#)

This 2003 article discusses how the role of admissions director has expanded to include marketing skills which have lead to the de-personalizing of recruitment and orientations, or “regression to a linguistic mean” (Donehower, 2003). The twist is that in this three paged analysis, there is a single, passing, paragraph that previews exactly the type of portal our group has researched, and is fast becoming the norm:

Although I have not yet seen a college Web site with the equivalent of an Amazon.com personalized welcome page for repeat visitors (“Hello, Nancy! See what’s new for you!”), a number of colleges collect the data that would make that possible. Many admissions offices monitor information obtained through e-mail, phone, and other contacts they have with applicants. (Donehower, 2003)

Assessment and Testing

Currently at Ohlone, the English and ESL departments are undergoing preparations to move to eAssessment going to the web-based test, Accuplacer. The primary motivation for this change was to offer to students’ instant results, and thereafter quick and accurate placement into classes. The test is still proctored, but in contrast to the prior test, which only showed results to the placement center, students can access their own information. It is felt that eventually connecting with Datatel, so that students may be instantly registered, is a worthwhile streamlining goal.

Career Planning

The goal of career planning is career advising and job search assistance for students who need to work while completing their studies. This is another offering in the eStudent Services and suite of services. The content includes: online office hours, job/internship postings online, online applications, job fairs, and recruitment of prospective employers. It also includes announcements of workshops and seminars to prepare students for careers after graduation.

There is also service like a Cooperative Education program, which offers students the opportunity to earn degree credit while working in a paid or volunteer position, related to their major or career goals.

In addition, online career planning hosts integrated onsite career planning services, as another way of assisting job search assistance.

Counseling

The purpose of online counseling services is to assist students in achieving educational goals. This service includes information on study skills test taking and how to succeed in college as well as information on mental health issues.

Interventions with students during crises or stress related to academic issues can lead to a lowered drop-out rate. Effective intervention through online services may lead to improved student retention.

Online Counseling services started with online learning. Counselors have been experimenting with online services since 1966 when an MIT professor created a program called “Eliza.”

Counseling information online provides students with access to sensitive information while they remain relatively anonymous. It allows them to access and meet staff online. Typically, online counseling services provide general information, counselor profiles and contact information.

Example of online counseling site

Columbia University

<http://www.goaskalice.columbia.edu/index.html>

Go Ask Alice provides an on line question and answer service related to the majority of mental health questions a college student might ask. It is a unique service for a college counseling web site.

Orientation

The goal of most student orientation program is to aid in matriculation, orientation, transition and retention of first year and transfer students. The

orientation must transmit a large amount of information to the students so that they understand the institution, learn the services, understand the academic requirements and develop a sense of community. This “information overload” can be delivered online in a very creative and personalized way, as a “one stop” for student support, by using personalized portals and resources, individualized checklists and links to specific academic resources. Online collaborations can include chat rooms, email, and discussion board, discussion groups with students that have the same program of study.

For some excellent examples of Online Student Orientations see:

University of Central Florida New Student Virtual Orientation

<http://reach.ucf.edu/~vorient/>

<http://learn.ucf.edu/orientation/index.html>

The University of Central Florida Virtual Orientation is designed for new transfer and distance students. Students move through a series of content pages and an associated quiz, which evaluates their understanding of the content. The student must obtain a perfect score on each quiz to proceed to the next area of content and course registration is not permitted until completing the full program.

University of Connecticut First Year Experience Program

<http://www.fye.uconn.edu>

The UConn FYE program has two online components. First, the FYE program is supplemented with online support, interaction, discussion and delivery of content and information. Secondly, the FYE portal provides first-year students with a range of student specific information including calendar of events, news, "ask the experts," "talk to your Orientation Leader" and discussion lists.

University of Utah New Student Orientation

https://www.saff.utah.edu/orient/programs/forms/form_other2.html

The University of Utah Online Orientation is designed for students who cannot come to campus for the face-to-face session. Broken into nine "puzzle pieces," the program introduces the student to campus resources, academic requirements, choosing classes and registering. As the student moves through the modules, they create a personal checklist that can be printed out as a list of reminders

Student Information System

The student information system is a student management tool, and is a part of the online student services that include student profile, scheduling, grade book,

attendance, report card, eligibility, transcripts, activities, class rank, gender, GPA, and more. The student can use this offering to keep track of schedule, student profile, address change, and more. The administrator can use this service to view student demographic, research, statistics, and other information.

Example: <http://www.ucd.ie/sisadmin/stuweb/>

Tutoring

The goal of tutoring is offering online tutoring services. When a student signs into the online student services web portal, he/she has access to email, virtual chat, discussion boards, self-help materials, writing centers, tutoring services, tutoring services or appointment services.

Projected Budget

How much would a portal cost to implement and support?

There is a very wide range of costs. Implementations using commercial products and consultants could run into **millions** of dollars. In addition, the overall systems cost require substantial dedicated effort, planning, communication and widespread participation across the institution. All will require ongoing support and content maintenance.

As with many IT projects, the biggest challenge is not the cost of the new technology, but the cost of continuing to support both the old and new technologies; the potential payoff is improved efficiency, organization, communication and sense of community.

PeopleSoft, Product Pricing Overview (2002),
<http://www.usdoj.gov/atr/cases/f204400/204465.htm>

“Contract vagueness, may be prevalent when a division of a large company wishes to buy a product and the tool figures can't be applied. For example, if a \$10B company wanted a CRM package for a division of 500 employees, the number of employees would not be solely counted and revenue would be too difficult to ascertain in a division. PeopleSoft will use a \$250,000 revenue number multiplied by the 500 employees as a factor, but the contract will say based.”

eStudent Services Roadmap

Resources like http://en.wikipedia.org/wiki/Web_portal give a reader a sense of the history and possible future for web portals. For instance, this site described how before the tech bust, it was thought that this tool would be widely used in the public sector (including a U.S. government web portal), but enthusiasm for their

use has waned. The suggestion was that business is again taking the lead in the sort of intranet technology and adaptation already mentioned in this report. These "enterprise web portals" are, in fact, cutting edge for education as seen in the recent merger/acquisition of WebCT/Blackboard which makes this functionality more widely available.

Extension Strategy Roadmap

Portal technologies and ERP have a wide range of applications and cross-functional areas. Future ERP for higher education may include functional departments that are involved in operations or production. By extension, if an institutional goal is to create a unified web interface for their students, faculty, administrators, and employees, then they will want to implement a student, faculty, administrator, and employee portal. Here is a list of applications that may be used with the web portal.

- Accounting
- Budgeting
- Human resources
- Marketing
- Strategic management
- Electronic Portfolios
- Blogs

Integration Roadmap

The eStudent Services is a web-based delivery of online student services (including both online-student services and online student-services). The portal architecture is a distributed and scalable system that deploys portal technology for the student, faculty, administrator, and employees. The integration is a complex process and it requires involvement from, and training of, individuals in many organizations such as: IT, institutional network systems, applications, eStudent Services, and more. Further analysis is required to determine areas impacted by integrating online student services and web portal technology.

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